



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

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Executive Summary

- SpaceY is a company that wishes to rival SpaceX in the space industry. In order to achieve this goal, SpaceY needs to determine the cost of SpaceX's launch. One of the metrics that determines this cost is the rate of successful and unsuccessful relaunches.
- A pivotal part of our analysis was the collection and cleaning of the data on SpaceX Falcon9 launches. This was used to identify the factors that gives rise to a successful mission outcome.
- We varied the parameters to determine the combination and model that will give the most accurate results. The modeling techniques used in the process are; Decision Tree, Logistic Regression, Smooth Vector Classification and K-Nearest Neighbor. And of these models, the most accurate is the Decision Tree.
- Our code and documentation can be found on Github.

Introduction

- SpaceX is the leading company in the rocket launch industry and SpaceY seeks to rival her.
- One of the reasons for the success had by SpaceX is its ability to offer lower prices for its launches. This is in part due to their ability to reuse the first stage of their rockets. This is the Falcon9 rocket.
- For SpaceY to compete with SpaceX, they need to determine when and how the first stage rocket lands.
- The questions that we seek to answer with this project are; *what are the overall costs of each rocket launch? What factors contribute to the success or failure of the SpaceX's first stage landing?*

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - The data were collected directly from SpaceX's API [here](#).
 - We also collected data via web-scraping the Wikipedia page for the Falcon9 Launches [here](#).
- Perform data wrangling
 - The data was cleaned to remove missing fields and develop indicator variables.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - We perform several techniques to build the model, with each of them possessing accuracy scores.

Data Collection

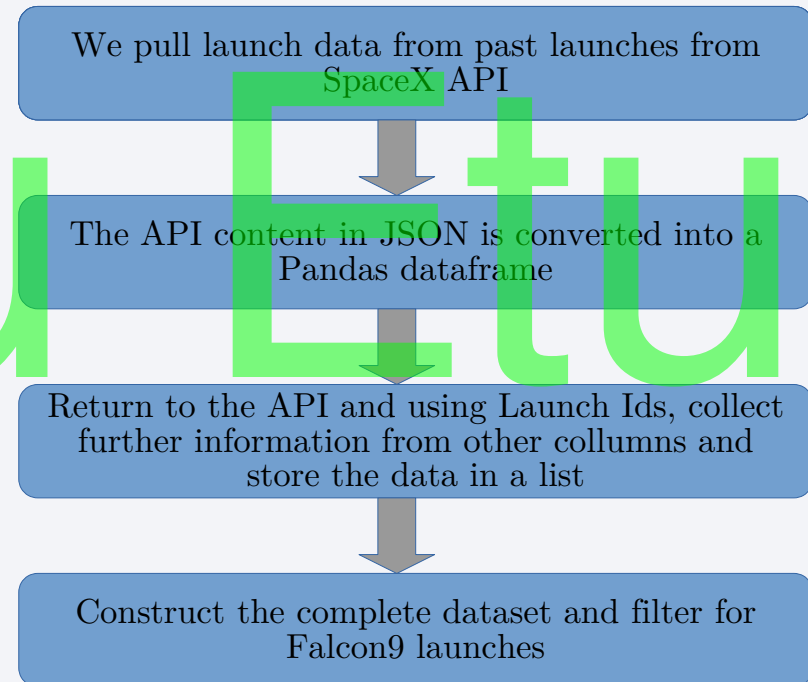
As previously said we used two methods to collect data on the Falcon9 launches, namely; The SpaceX API and from the Wikipedia page of Falcon9.

Other variables collected using the above methods are: *Flight number, Outcome, Data, Booster version, Payload mass, Orbit, Launch Site, Flights, Reused, Grin fins, Landing pads, Legs, Serial, Block, Reused count, Longitude, Latitude.*

Data Collection – SpaceX API

- We collected data directly from SpaceX's API using REST calls.
- Leveraging on information from specific columns such as; *Launchpad*, *Rocket*, *Payload* and *Cores* columns, we access the API.
- **Notebook 1** contains the Jupyter notebook on the data collection API.

Data Collection Process using SpaceX's API



Data Collection - Scraping

- Subsequently, we collected data directly from Wikipedia from SpaceX's Falcon9 Launches.
- Notebook 2 contains the Jupyter notebook for the web-scraping process.

Data Collection Process via Web-scraping

Request the Falcon9 launch Wikipedia page from its URL.

Use the HTTP response to create a BeautifulSoup Library.

Extract all column names from the HTML table header.

Create a Pandas dataframes by parsing the launch HTML Tables.

Data Wrangling

- The data collection process resulted in a dataset with 17 variables.
- Some variables that were of particular interest after exploration were: *The launch sites and the possible orbits.*
- For a less de-cluttered system of analysis we assign the value 1 to a successful outcome and the value 0 to the failure outcome.
- Notebook 3 contains the Jupyter notebook for the data wrangling portion of this project.

EDA with Data Visualization

- Using Pandas and Matplotlib libraries, we conducted Exploratory Data Analysis (EDA), visualizing the relationships between different parameters, such as; *Launch site by payload mass, Orbit type payload mass, Payload by flight number, Success rate by orbit type, e.t.c.*
- Notebook 5 contains the EDA conducted with the data visualization techniques.

EDA with SQL

- EDA was performed for SQL queries.
- The following were examined to search for patterns in the data: *Unique launch sites, Total payload mass carried by boosters launched by NASA, Average payload mass carried by booster version F9 v1.1, List of failed launches between 4th June 2010 and 20th March 2017, etc.*
- Notebook 4 contains the code for EDA using SQL.

Build an Interactive Map with Folium

- We conducted a geographical analysis of the launch sites using the Folium library.
- These helped greatly in the analyses between the success rates of the launches and the launch sites.
- Notebook 6 contains Jupyter notebook of the interactive maps created with Folium.

Build a Dashboard with Plotly Dash

- Using the Plotly and Dash Python libraries, a SpaceX records dashboard was created.
- Two plots was used to display data on all launch sites. These give in an interactive way the visualizations of the success rates, filtering them by launch sites. This allows for patterns to be observed by launch sites and payload sizes for launches.
- Python script of the completed Plotly Dash lab code.

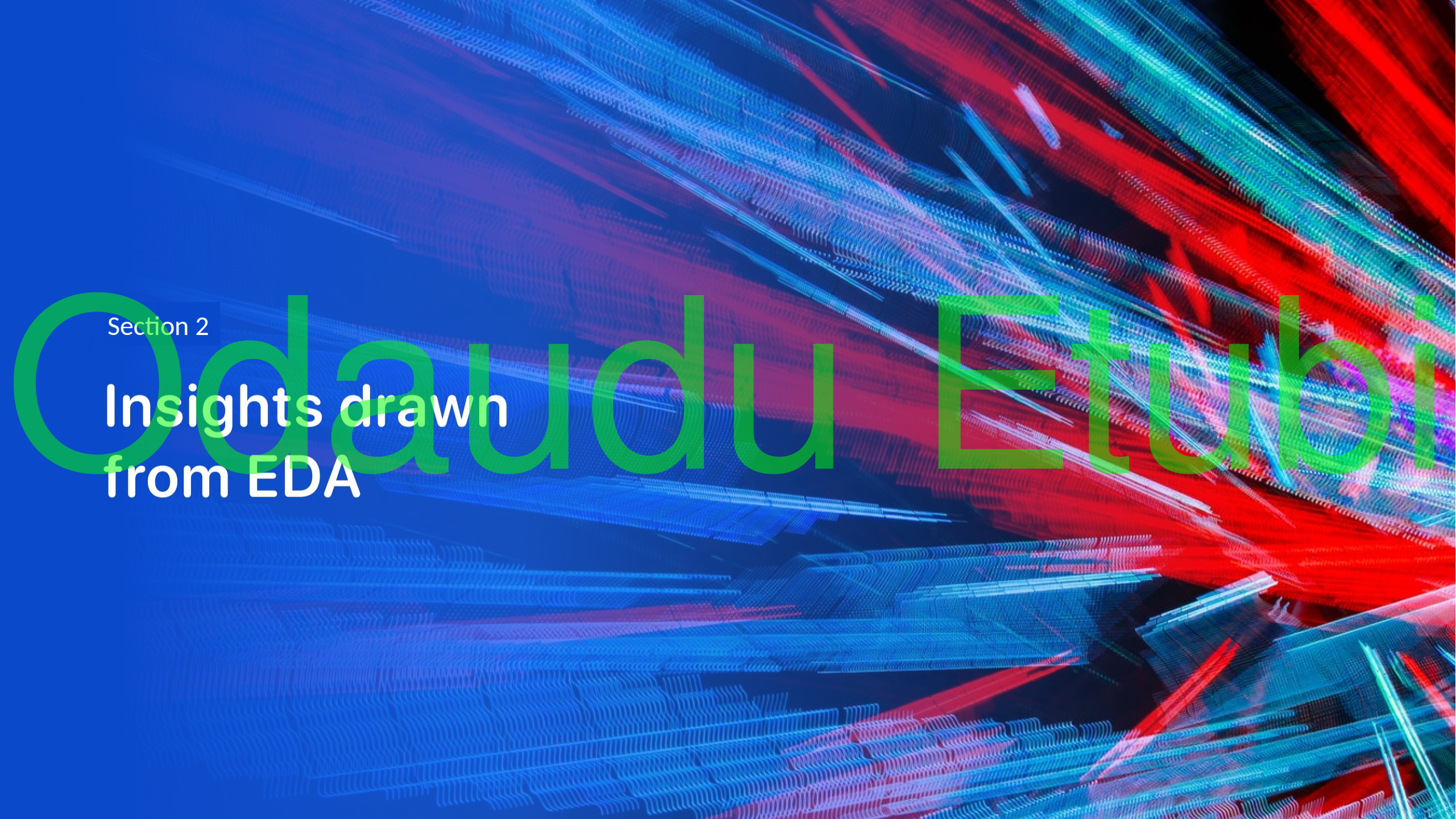
Predictive Analysis (Classification)

- Multiple modeling techniques and strategies were explored to determine the best predictor of the success and failure of launches. The process followed a standard flow which consisted of; *Preprocessing the data before splitting it into training and testing sets, Using models to test our data, Collect information to assess model performance.*
- Notebook 8 contains the Jupyter notebook code for the predictive analysis.

Results

- Exploratory data analysis results were obtained
- Interactive analytics were gotten using Folium.
- Our best model technique gives an accuracy of 83.3% score

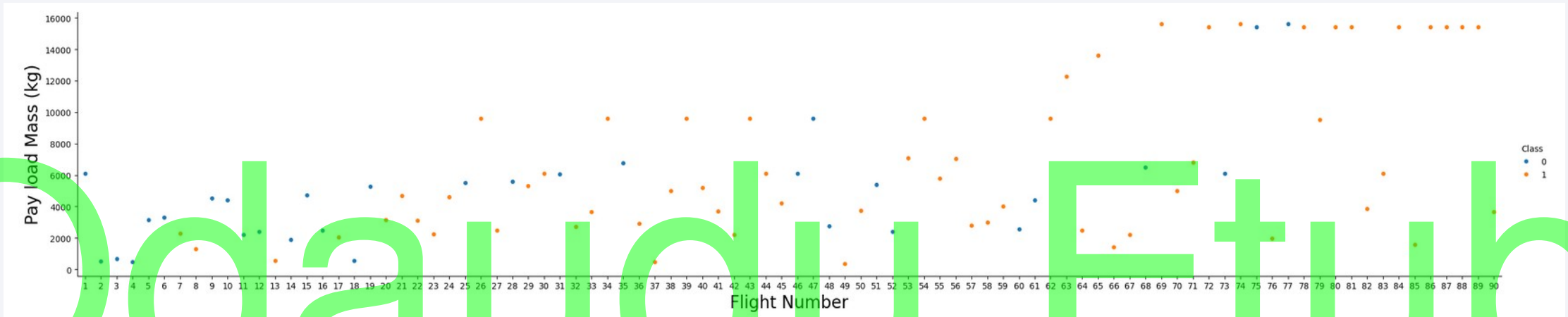
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The background is a dynamic, abstract composition of numerous thin, overlapping lines and streaks in shades of blue and red, creating a sense of motion and depth. A large, semi-transparent green watermark with the text 'Odaudu Etubi' is overlaid across the center of the image.

Section 2

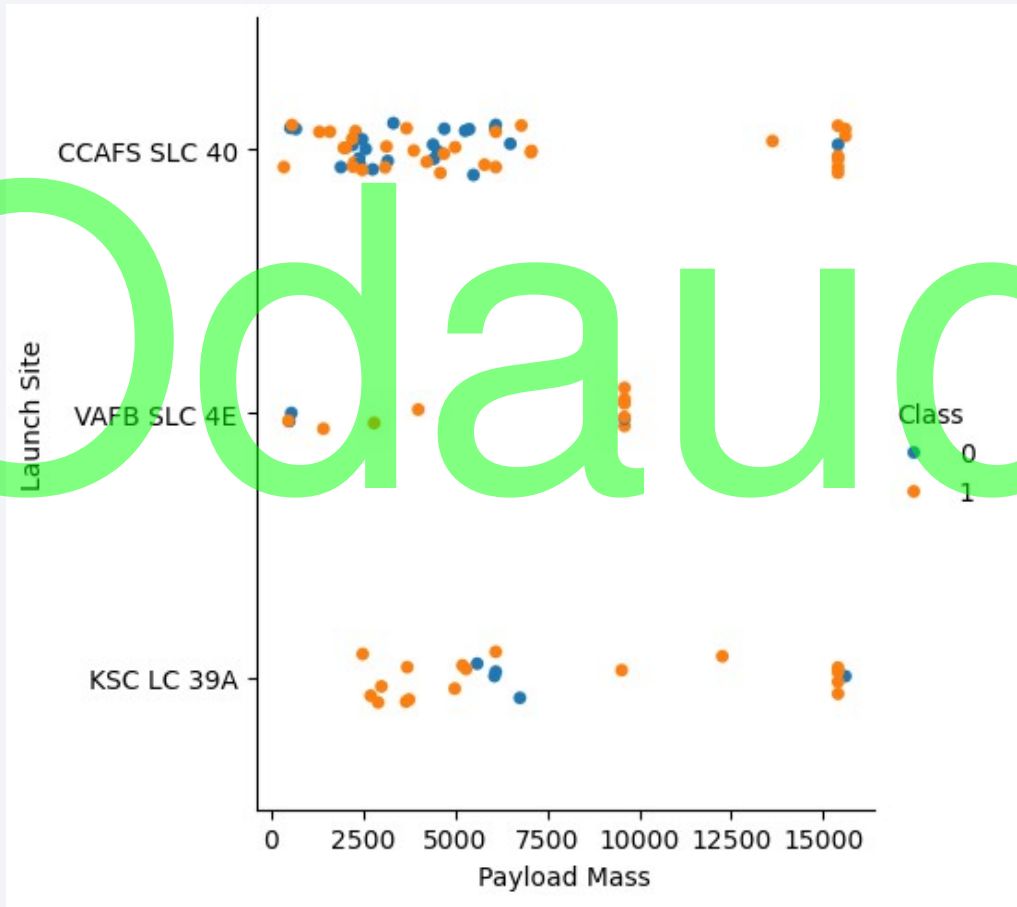
Insights drawn
from EDA

Flight Number vs. Launch Site



- The plot above shows the payload on the vertical axis and the flight number on the horizontal axis. It clearly depicts the increase in the maximum payload over time, with a steady increase until we hit the threshold of 10,000kg at the flight 70.
- There is a steady increase of the orange dots (successful launches) over the blue dots (failed launches), which signals progress over time.

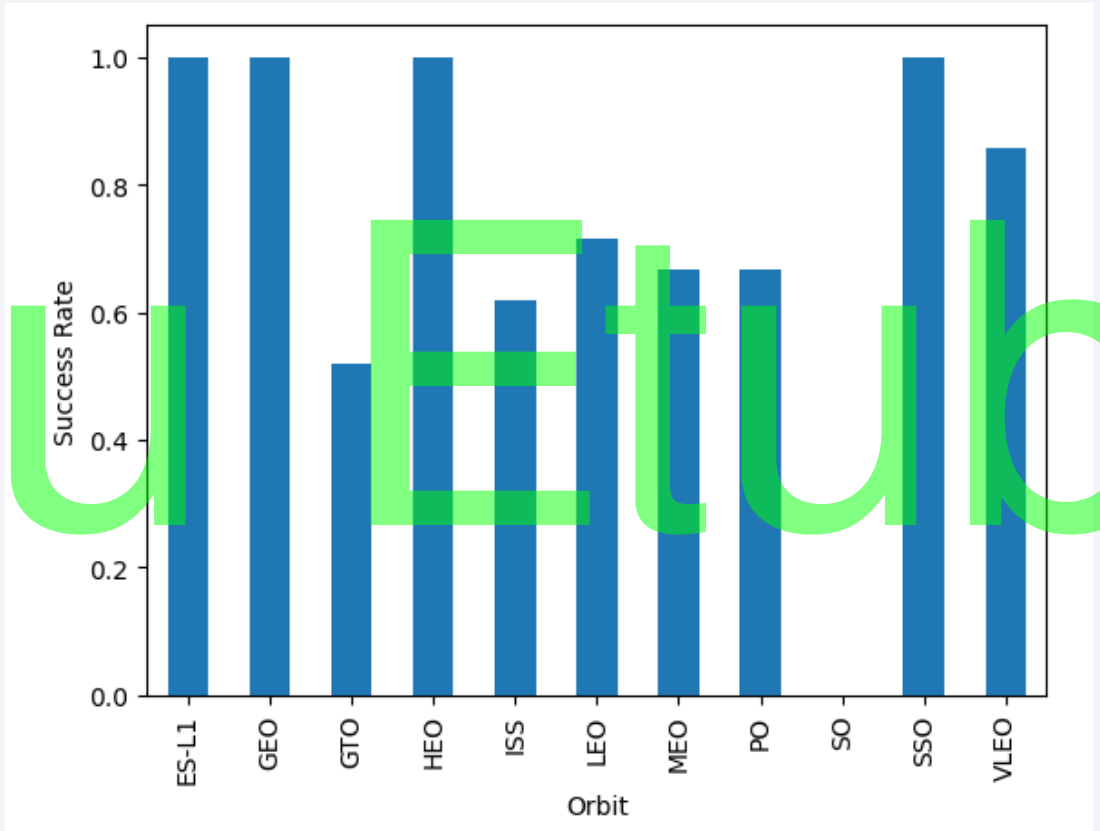
Payload vs. Launch Site



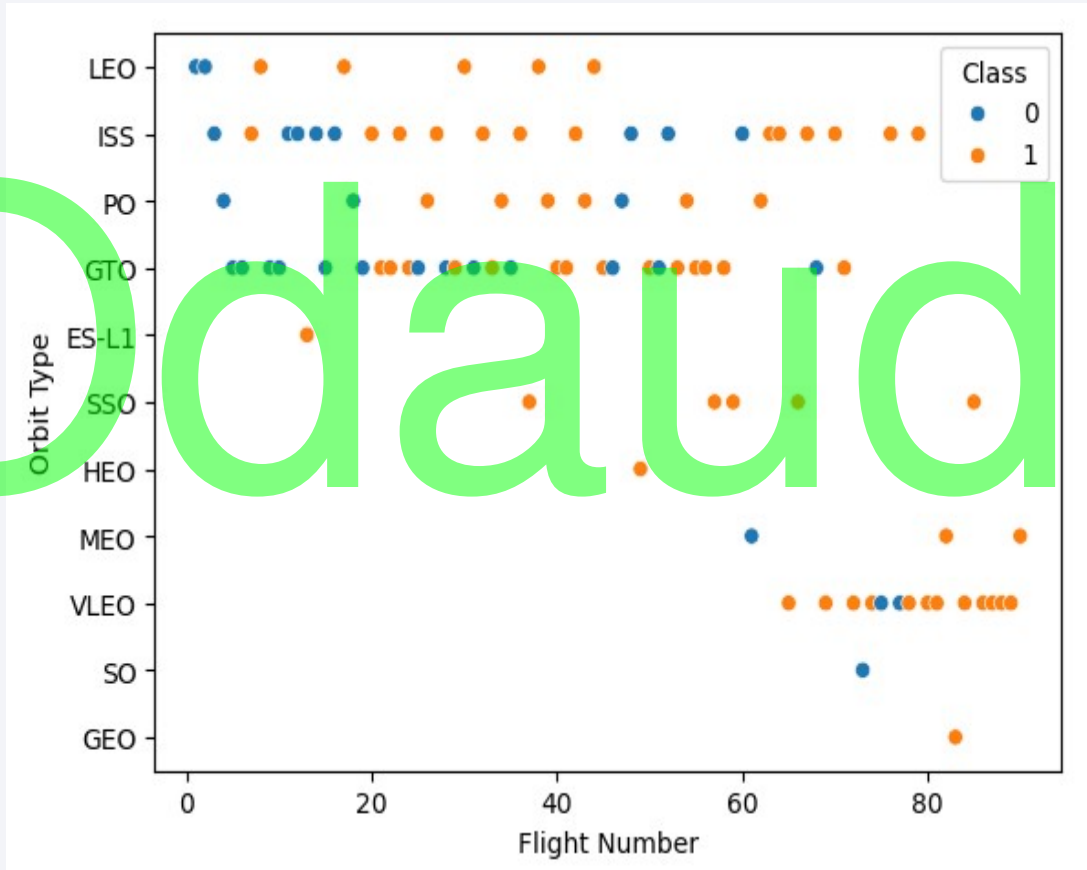
- From the plot we see that the CCAFS SLC 40 has the most launches, by the frequency of dots (orange and blue) it has.
- On average, VAFB SLC 4E has trended towards lower payload sizes.
- The 3 sites have a mix of results, where high payloads tend to have high success rates.

Success Rate vs. Orbit Type

- The success rates are much higher for orbits that are farther from the earth.
- From the graph on the right, we see the relationship between the orbit type and success rate.



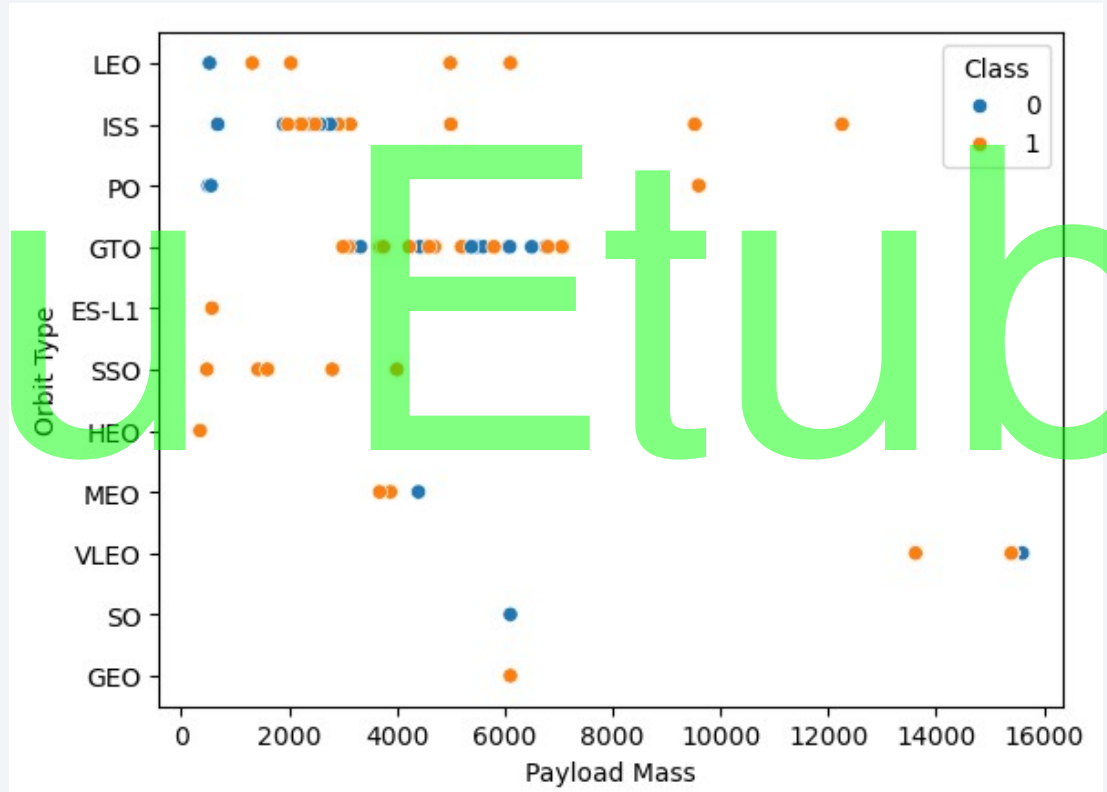
Flight Number vs. Orbit Type



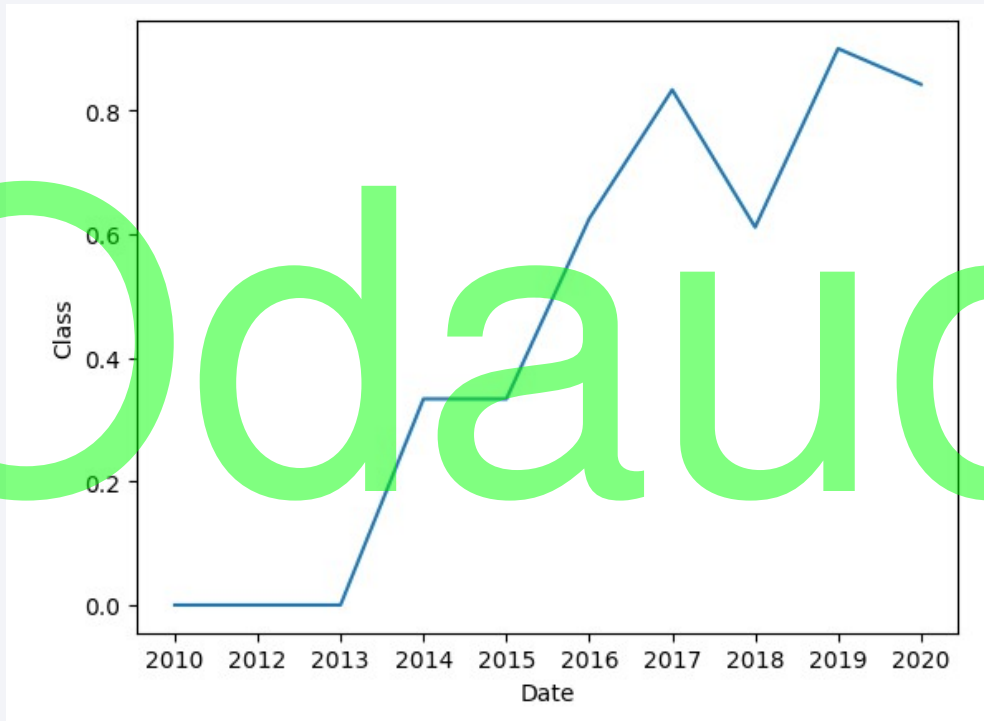
- SpaceX has attempted new orbit types as depicted in the graph.
- The relationship between these two variables doesn't seem so strong.

Payload vs. Orbit Type

- Payload mass affect which orbits are attempted.
- As can be seen, VLEO is only attempted for payloads greater than 13000kg. Also, ISS orbits has the most varied payload range.
- Payload mass determines the kinds of orbits attempted.



Launch Success Yearly Trend



- There have been failed landings between 2010 and 2013.
- Successful launches began from 2014 onwards.
- From the graph, we observe that SpaceX has had improved annual success over time.

All Launch Site Names

- This SQL query extracts a list of unique launch site values.
- From the 4 launch sites depicted, 2 of them are actually Cape Canaveral. The CCAFS LC-40 and the CCAFS SLC-40.

```
%%sql
--select * from SPACEXTABLE LIMIT 5
select distinct Launch_Site from SPACEXTABLE
```

* sqlite:///my_data1.db
Done.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%%sql
select * from SPACEXTABLE
where Launch_Site like 'CCA%'
limit 5
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

This SQL query gives the first 5 launches that took place at Cape Canaveral.

Total Payload Mass

```
%%sql
select sum(PAYLOAD_MASS_KG_) as Nasa_CRS_payload_sum from SPACEXTABLE
where Customer = 'NASA (CRS)'

* sqlite:///my_data1.db
Done.
: Nasa_CRS_payload_sum
-----
45596
```

This SQL query calculates the total payload mass launched by NASA over all launches.

Average Payload Mass by F9 v1.1

```
%%sql
select avg(PAYLOAD_MASS_KG_) as f9_1v1_payload_avg from SPACEXTABLE
where Booster_Version like 'F9 v1.1%'

* sqlite:///my_data1.db
Done.
  f9_1v1_payload_avg
-----
2534.6666666666665
```

This SQL query calculates the average payload for F9 v1.1

First Successful Ground Landing Date

```
%%sql  
select min(Date) from SPACEXTABLE  
where Landing_Outcome = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db
```

```
Done.
```

min(Date)

2015-12-22

This SQL query returns the date of the first successful landing on a ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
--select distinct Landing_Outcome from SPACEXTABLE

select Booster_Version from SPACEXTABLE
where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS_KG_ > 4000 and PAYLOAD_MASS_KG_ < 6000

* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

This SQL query returns a list of F9 rockets that have successfully landed on a drone ship carrying payloads between 4000kg and 6000kg.

Total Number of Successful and Failure Mission Outcomes

```
%%sql
--select distinct Mission_Outcome from SPACEXTABLE

select distinct
  (select count(Mission_Outcome) from SPACEXTABLE
   where Mission_Outcome Like '%Success%') as success_count,
  (select count(Mission_Outcome) from SPACEXTABLE
   where Mission_Outcome Like '%failure%') as failure_Count
FROM SPACEXTABLE

* sqlite:///my_data1.db
Done.
```

success_count	failure_Count
100	1

An SQL query that returns a total count of all mission outcomes.

Boosters Carried Maximum Payload

This SQL query returns a list of F9 that have ever carried the maximum payload.

```
%sql
select distinct Booster_Version from SPACEXTABLE
where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)

* sqlite:///my_data1.db
Done.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7
```

2015 Launch Records

```
%%sql
select substr(Date, 6, 2) as month,
substr(Date, 0, 5) as year, Booster_Version, Launch_Site from SPACEXTABLE where
landing_outcome = "Failure (drone ship)" and year = "2015"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

month	year	Booster_Version	Launch_Site
01	2015	F9 v1.1 B1012	CCAFS LC-40
04	2015	F9 v1.1 B1015	CCAFS LC-40

This SQL query returns a list of all 2015 launch records with unsuccessful landings on a drone ship, including launch site and version of the F9 booster used.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
select launch_site, count(Mission_Outcome) as MISSION_OUTCOME_COUNT
from SPACEXTABLE
where date between "2010-06-04" and "2017-03-20"
group by Launch_Site
order by MISSION_OUTCOME_COUNT DESC

* sqlite:///my_data1.db
Done.
```

Launch_Site	MISSION_OUTCOME_COUNT
CCAFS LC-40	26
VAFB SLC-4E	3
KSC LC-39A	2

This SQL query ranks the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order



Section 3

Launch Sites Proximities Analysis

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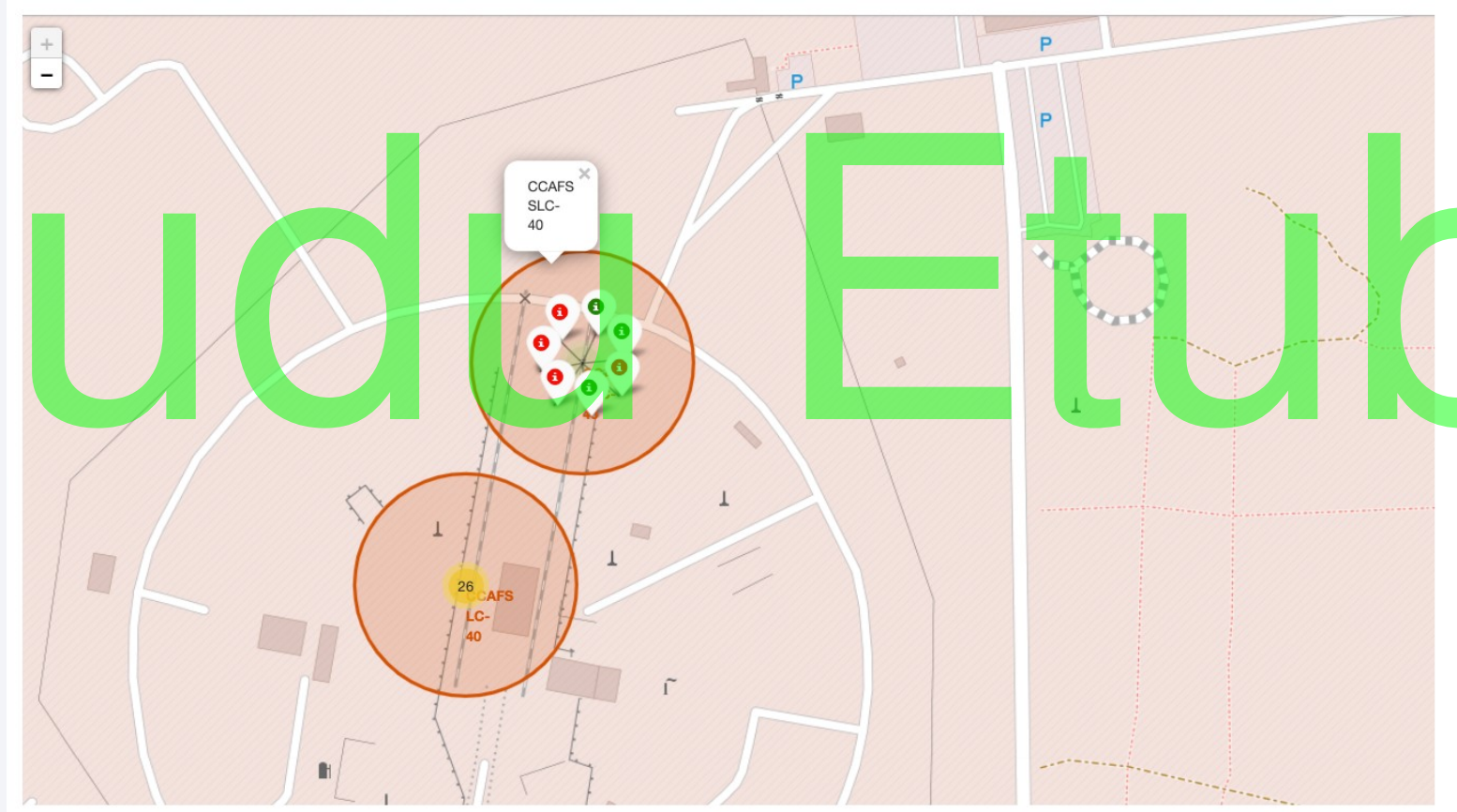
Folium Map showing Key locations of SpaceX Launches

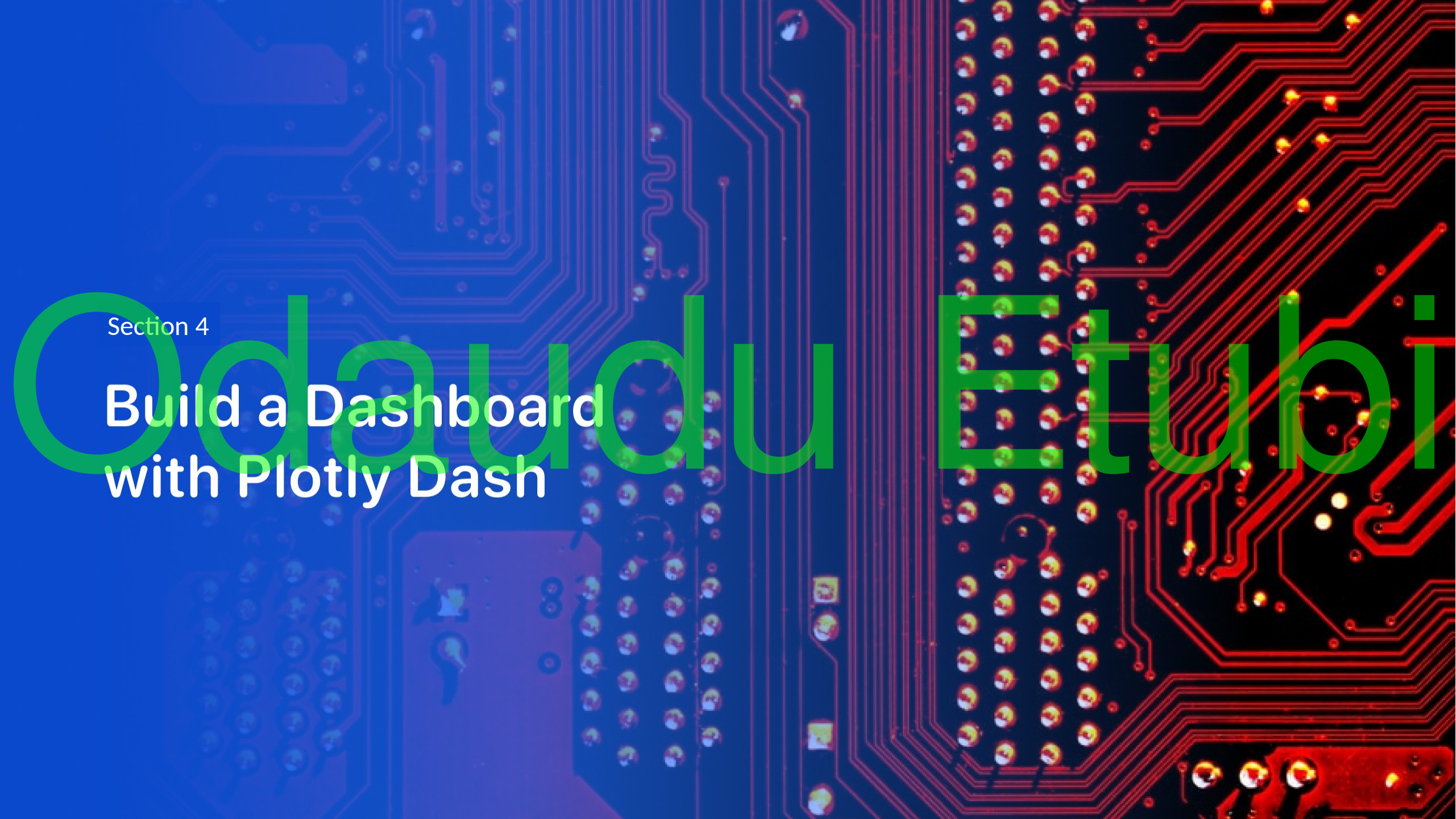


- All launch sites are in the southern part of the USA.
- The highest amount of launches have been in Florida.

Launches for Individual Rockets.

The Folium map showing individual launches.



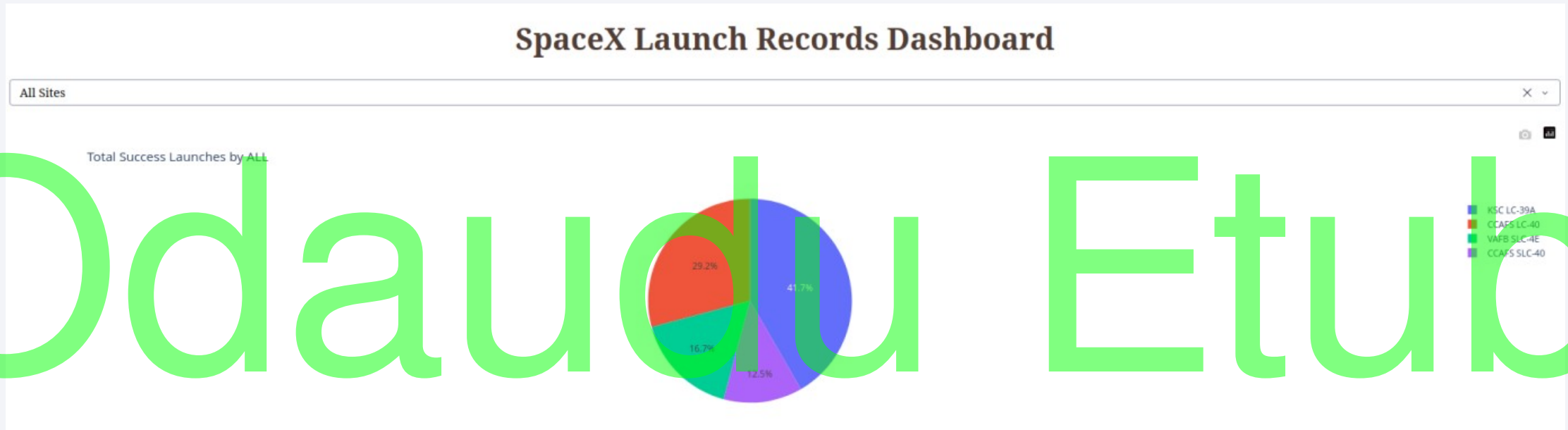


Section 4

Build a Dashboard with Plotly Dash

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Dashboard for the launch sites success rates



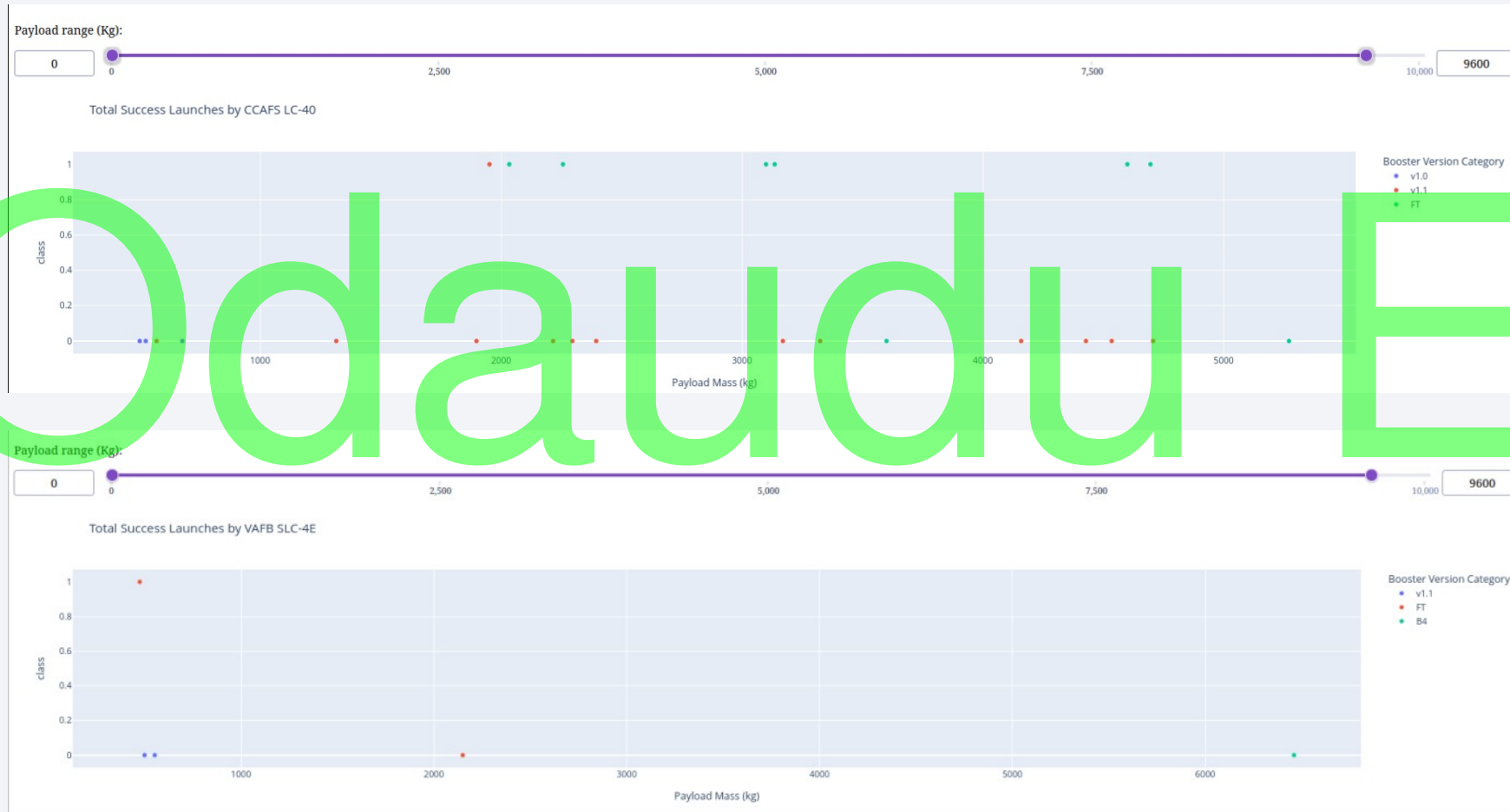
- The success rates from Central Florida occurred at Cape Canaveral and the Kennedy Space Center. And these accounted for most of the success.
- Vandenberg Space Force Base is responsible for about 17% of successful launches.

A filter showing the success rate of an individual site



- The success rate with the highest launch ratio of all launch sites is the Vandenberg Space Force Base at about 77%.

A scatter plot maps successes



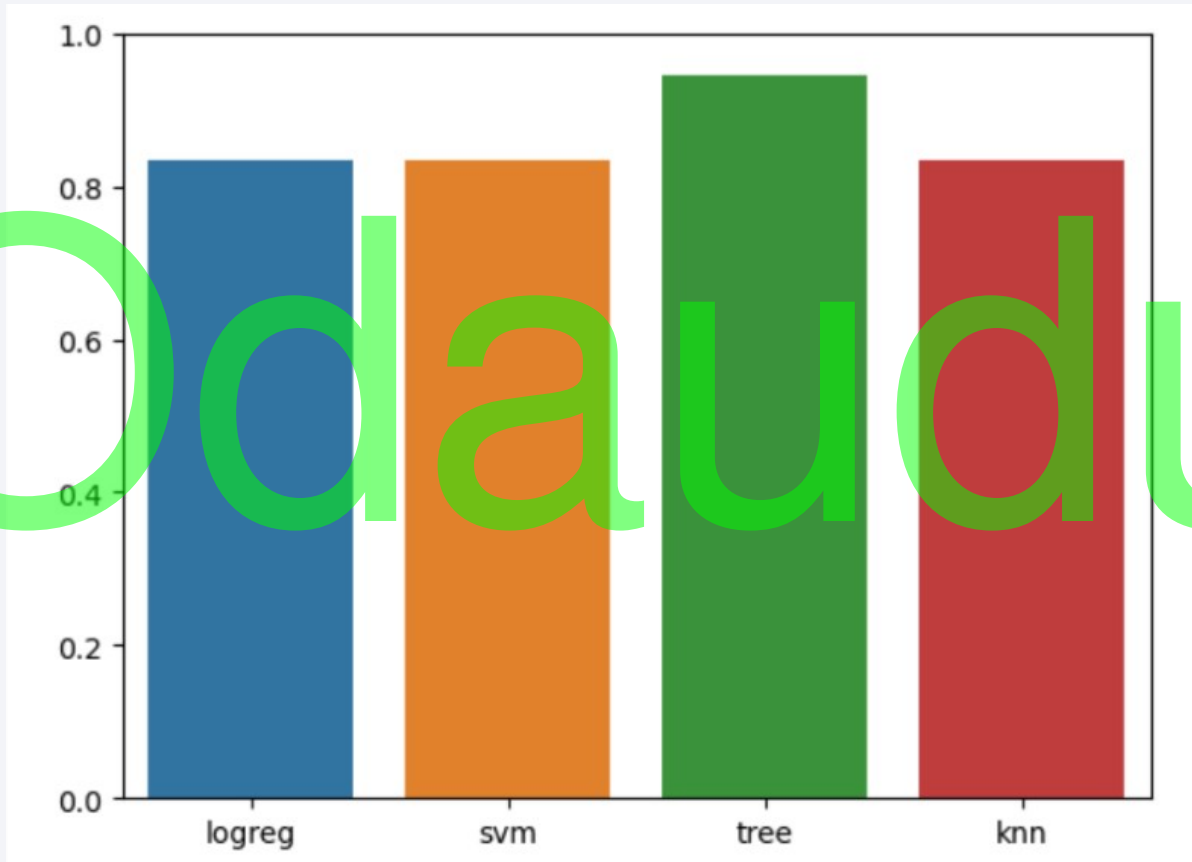
- A plot showing selected launch sites.
- The graph is shown when the payload is set at 0.
- The scatter plot changes as we slide the payload.

Section 5

Predictive Analysis (Classification)

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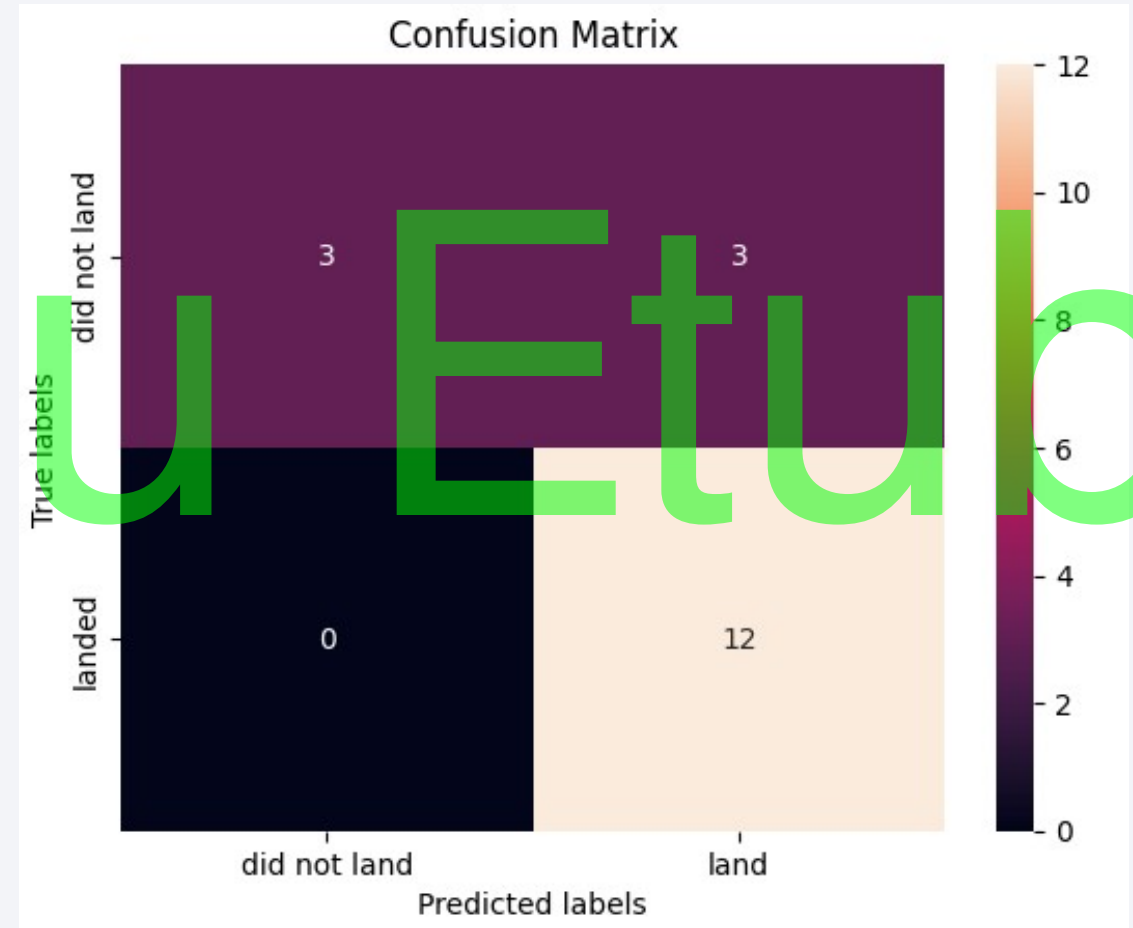
Classification Accuracy



- We calculate accuracy scores were calculated for each of the 4 model types.
- From the graph, the highest accuracy score at 94% from the Decision Tree modeling technique.

Confusion Matrix

- The confusion tree has only one false positive.



Conclusions

Some of the things we discovered from our analysis were that;

- Some intricate relationships between the successes of a launch and its payload, orbit and launch sites.
- Launch sites were always located near major landmarks.
- The Decision tree model technique gave us the highest accuracy score.



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Thank you!